

2. A line AB of 70 mm length has its end A 10 mm above H.P. and 15 mm in front of V.P. The other end is 40 mm above H.P. and 60 mm in front of V.P. Draw its projections and determine its inclination with H.P. and V.P.

Solution : Refer fig. 5.32.

Procedure :

1. Since A is 10 mm above H.P.; mark a' 10 mm above x-y line.
2. Since A is 15 mm in front of V.P.; mark a 15 mm below x-y line.
3. Since B is 40 mm above H.P., draw locus of b' 40 mm above x-y line.
4. Since B is 60 mm in front of V.P., draw locus of b 60 mm below x-y line.

To obtain θ and ϕ :

5. Since line is 70 mm long; with center a' and radius = T.L. = 70 mm, cut the locus of b' at b_1' . Join $a'-b_1'$. Measure θ .
6. Now, with center a and radius = T.L. = 70 mm, cut the locus of b at b_2 . Join $a-b_2$. Measure ϕ .

To draw F.V. and T.V. :

7. Project b_1' vertically downward on the horizontal line through a and mark b_1 . With center a and radius = $a-b_1$, draw an arc to cut the locus of b at b . Join $a-b$, which is actual T.V. of line AB.
8. Project b_2 vertically upward on the horizontal line through a' and mark b_2' . With center a' and radius = $a'-b_2'$, draw an arc to cut the locus of b' at b' . Join $a'-b'$, which is actual F.V. of line AB.

Check : Points b' and b lies on the same projector.

Given

AB = 70 mm

A is 10 mm above H.P.

A is 15 mm in front of V.P.

B is 40 mm above H.P.

B is 60 mm in front of V.P.

Conclusions

→ T.L. = $a'-b_1' = a-b_2 = 70$ mm

→ a' is 10 mm above x-y line

→ a is 15 mm below x-y line

→ b' is 40 mm above x-y line

→ b is 60 mm below x-y line

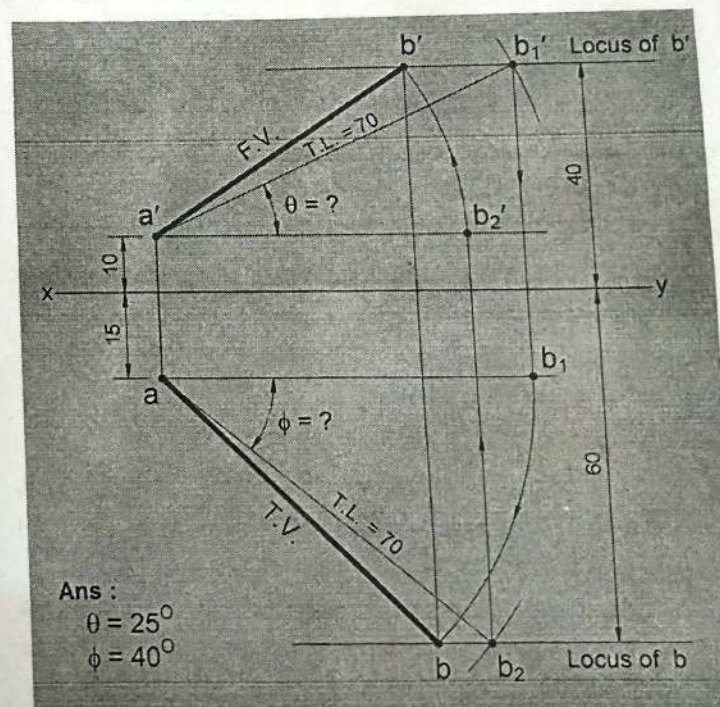


Fig. 5.32

6. The top view of a 75 mm long line AB measures 65 mm, while the length of its front view is 50 mm. Its one end A is in the H.P. and 12 mm in front of the V.P. Draw the projections of AB and determine its inclinations with the H.P. and the V.P.

Dr.B.A.T.U. [May 2017]

Solution : Refer fig. 5.36.

Procedure :

1. Since A is in H.P.; mark a' on x-y line.
2. Since A is 12 mm in front of V.P.; mark a 12 mm below x-y line.
3. Since F.V. of line measures 50 mm; through a' , draw $a'-b_2' = 50$ mm along x-y line. Draw vertical projector through b_2' .
4. Since T.V. of line measures 65 mm; through a , draw $a-b_1 = 65$ mm $\parallel$ to x-y line. Draw vertical projector through b_1 .
5. Since T.L. of line is 75 mm; with center a' and radius = 75 mm cut the projector drawn through b_1 at b_1' . Join $a'-b_1'$. Measure θ . Draw locus of b' through b_1' . [Cut the projector of b_1 above the x-y line, assuming that B is in 1st quadrant]
6. Since T.L. of line is 75 mm; with center a and radius = 75 mm cut the projector drawn through b_2' at b_2 . Join $a-b_2$. Measure ϕ . Draw locus of b through b_2 . [Cut the projector of b_2' below the x-y line, assuming that B is in 1st quadrant]
7. With center a' and radius = $a'-b_2'$, draw an arc to cut the locus of b' at b' . Join $a'-b'$. $a'-b'$ is the F.V. of line.
8. With center a and radius = $a-b_1$, draw an arc to cut the locus of b at b . Join $a-b$. $a-b$ is the T.V. of line.

Check : Points b' and b lies on the same projector.

Given data

AB = 75 mm

A is in H.P.

A is 12 mm in front of V.P.

F.V. measures 50 mm

T.V. measures 65 mm

Conclusions

$\rightarrow$ T.L. = $a'-b_1' = a-b_2 = 75$ mm

$\rightarrow$ a' is on x-y line

$\rightarrow$ a is 12 mm below x-y line

$\rightarrow$ $a'-b' = a'-b_2' = 50$ mm

$\rightarrow$ $a-b = a-b_1 = 65$ mm

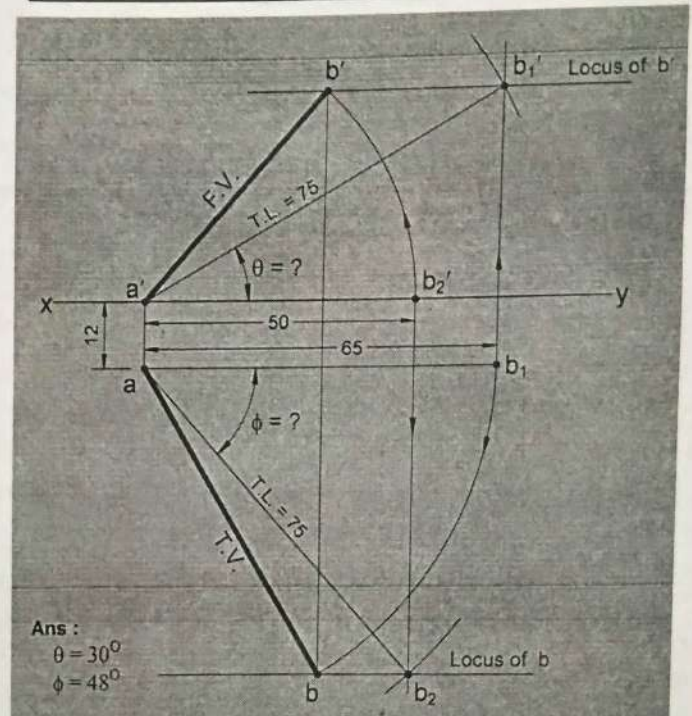


Fig. 5.36

10. The distance between the end projectors of a line AB is 40 mm. The end A is 10 mm above H.P. and 20 mm in front of V.P. The end B is 45 mm above H.P. and 60 mm in front of V.P. Draw the projections of line and find its true length and inclination with H.P. and V.P.

Solution : Refer fig. 5.40.

Procedure :

1. Since distance between end projectors is 40 mm; draw projector of A and B 40 mm apart.
2. Since A is 10 mm above H.P. and 15 mm in front of V.P.; on projector of A, mark a' 10 mm above x-y line and mark a 15 mm below x-y line.
3. Since B is 45 mm above H.P. and 60 mm in front of V.P.; on projector of B, mark b' 45 mm above x-y line and mark b 60 mm below x-y line.
4. Join $a'-b'$, which is F.V. of the line AB. Through b' , draw locus of b' horizontal. Join $a-b$, which is T.V. of the line AB. Through b , draw locus of b horizontal.

To obtain T.L, θ and ϕ :

5. With center a' and radius = $a'-b'$, draw an arc to cut the horizontal line through a' at b_2' . Project b_2' vertically downward on locus of b and mark b_2 . Join $a-b_2$. Measure T.L and ϕ .
6. Similarly, with center a , radius = $a-b$, draw an arc to cut the horizontal line through a at b_1 . Project b_1 vertically upward on locus of b' and mark b_1' . Join $a'-b_1'$. Measure T.L and θ .

Check : True Length obtained $a'-b_1'$ = True Length obtained $a-b_2$.

Given

Conclusions

Distance between projector of A and B = 40 mm

A is 10 mm above H.P. $\rightarrow a'$ is 10 mm above x-y line
 A is 15 mm in front of V.P. $\rightarrow a$ is 15 mm below x-y line
 B is 45 mm above H.P. $\rightarrow b'$ is 45 mm above x-y line
 B is 60 mm in front of V.P. $\rightarrow b$ is 60 mm below x-y line

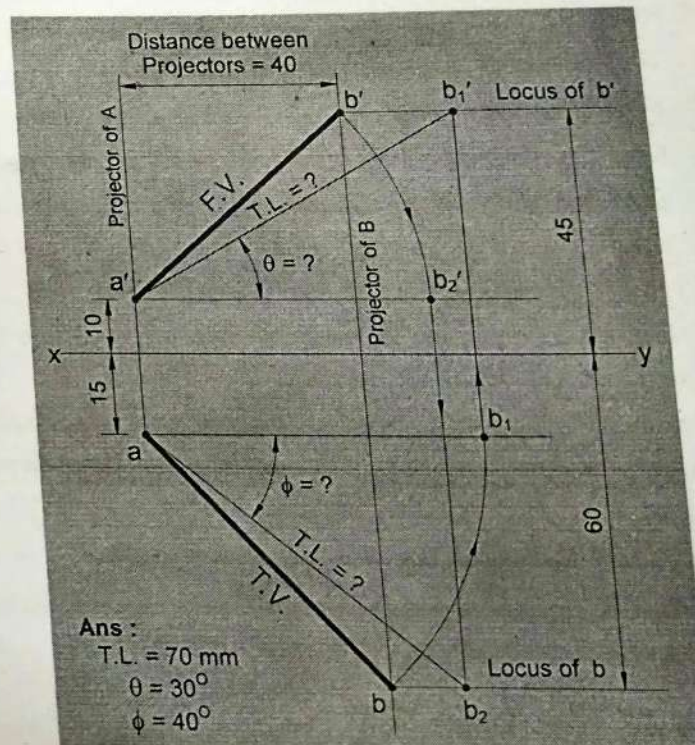


Fig. 5.40

16. Draw the projections of line "AB", if Point "A" is 20 mm above H.P. and 30 mm in front of V.P. Point "B" is 75 mm above H.P. and 60 mm in front of V.P. and distance between projectors of AB is 60 mm. Find H.T., V.T., True Length of line and Angle made with H.P.

12 Marks, Dr.B.A.T.U. [Dec. 2017]

Solution : Refer fig. 5.46.

Procedure :

1. Since distance between end projectors of AB is 60 mm; draw projector of A and B 60 mm apart.
2. Since A is 20 mm above H.P. and 30 mm in front of V.P.; on projector of A, mark a' 20 mm above x-y line and mark a 30 mm below x-y line.
3. Since B is 75 mm above H.P. and 60 mm in front of V.P.; on projector of B, mark b' 75 mm above x-y line and mark b 60 mm below x-y line.
4. Join $a'-b'$, which is F.V. of the line AB. Through b' , draw locus of b' horizontal. Join $a-b$, which is T.V. of the line AB. Through b , draw locus of b horizontal.

To obtain T.L, θ and ϕ :

5. With center a' and radius = $a'-b'$, draw an arc to cut the horizontal line through a' at b_2' . Project b_2' vertically downward on locus of b and mark b_2 . Join $a-b_2$. Measure T.L and ϕ .
6. Similarly, with center a , radius = $a-b$, draw an arc to cut the horizontal line through a at b_1 . Project b_1 vertically upward on locus of b' and mark b_1' . Join $a'-b_1'$. Measure T.L and θ .

Check : True Length obtained $a'-b_1'$ = True Length obtained $a-b_2$.

Traces : Mark h , where extension of $a'-b'$ cuts x-y line. Mark H.T., where projector through h cuts the extension of $a-b$. Mark v , where $a-b$ cuts x-y line. Mark V.T., where projector through v cuts $a'-b'$.

Given

Distance between projector of A and B = 60 mm

A is 20 mm above H.P.

A is 30 mm in front of V.P.

B is 75 mm above H.P.

B is 60 mm in front of V.P.

Conclusions

$\rightarrow a'$ is 20 mm above x-y line

$\rightarrow a$ is 30 mm below x-y line

$\rightarrow b'$ is 75 mm above x-y line

$\rightarrow b$ is 60 mm below x-y line

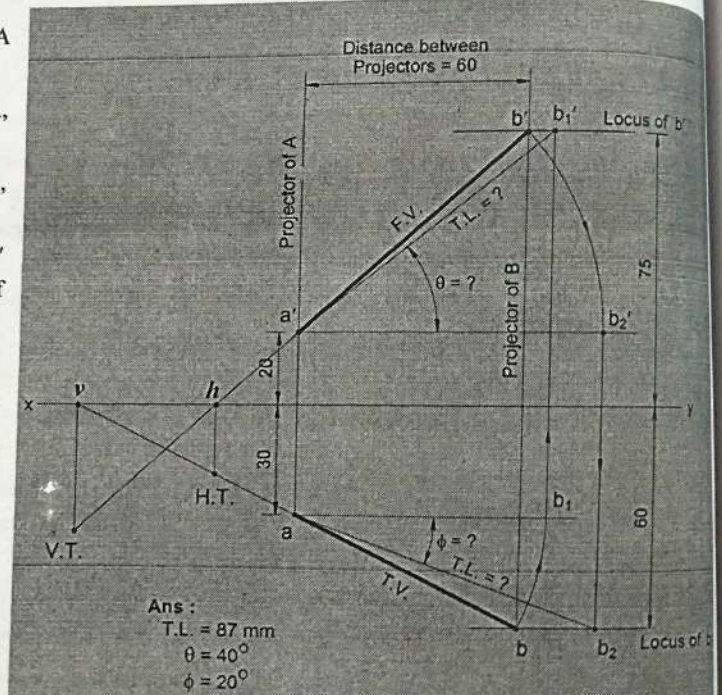


Fig. 5.46